

Intent Corruption: When AI Writing Tools Quietly Change What You Meant

A Taxonomy, a Benchmark, and a Detector for Pragmatic Drift in AI-Mediated Text

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With thanks to Ruchitha, whose conversation began this work.

<https://github.com/KrishnaShettyDev/ruflo-sil.git>

Abstract

AI writing tools change what users mean. Sometimes they flip a negation. Sometimes they swap a name. Sometimes they soften a refusal into an agreement across multiple hops of an agent pipeline. We call this phenomenon *intent corruption*. The field has lacked a name for it, a taxonomy describing it, a benchmark to measure it, or a detector to catch it. This paper provides all four. We define a twelve-category taxonomy (ICOR) grounded in pragmatic linguistic theory. We build SIL (Semantic Intent Layer), a detector ensemble that combines rule-based filters, a natural language inference cross-encoder, and a large language model judge. On a 1,460-example synthetic benchmark (ICOR-Bench) SIL reaches F1 of 93.9 percent. On 1,572 human-annotated revisions from the IteraTeR corpus, F1 drops to 67.6 percent with Cohen's kappa of 0.453, indicating moderate agreement with human ground truth. The 26-point F1 gap between in-distribution and naturalistic evaluation is the paper's most important finding. We diagnose two mechanisms behind the gap: the NLI component over-fires on sentence reordering, and the category-specific detectors under-fire while a generic embedding-divergence signal absorbs most of the true positives. We argue this gap is structural to the field, not unique to our system, and outline architectural fixes as the natural next step.

1. A Forty-Seven Minute Misunderstanding

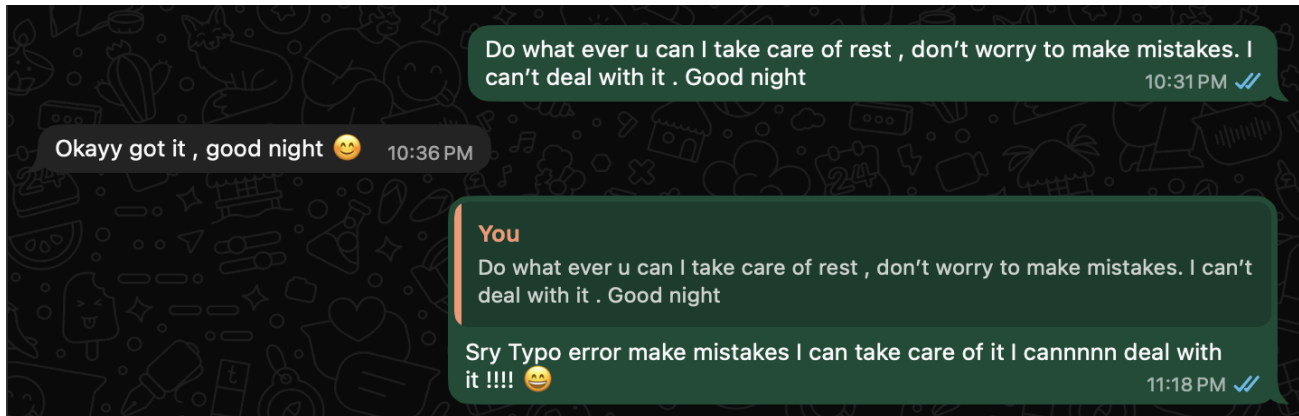
On the evening of May 17, 2026, the lead researcher at Plutas Lab sent a WhatsApp message to a friend named Ruchitha. The message was meant to say *do whatever you can, I will take care of the rest, don't worry about making mistakes, I can deal with it*. What actually got sent was: *Do what ever u can I take care of rest, don't worry to make mistakes. I can't deal with it. Good night*.

Ruchitha read the message as written. She replied okay, good night. Forty-seven minutes later the sender re-read the conversation, noticed the typo, and sent a correction. For forty-seven minutes one person believed the opposite of what the other one meant.

The cause was not an artificial intelligence system. It was a phone keyboard inserting an apostrophe and a letter t where none belonged. The mechanism is mundane. The consequence is not. A single character flip turned a reassuring message into a refusal, and the recipient had no way to know it had happened.

[Figure 1: WhatsApp screenshot to be inserted by the author. Caption below.]

Figure 1. A WhatsApp exchange (May 17, 2026, 10:31 PM IST). The sender's typo "I can't deal with it" inverted the intended meaning. The recipient replied at 10:36 PM. The sender noticed and corrected the mistake at 11:18 PM. Image included with the friend's explicit written permission.



We open with this anecdote because it makes the problem concrete in a way that abstract examples do not. A single token flipped, in a real conversation, between real people, with real social consequences. The keyboard did not warn anyone. The recipient could not have known the message was wrong without comparing it against the sender's mental model. There was no detection layer in the loop.

The rest of this paper argues that this exact failure mode occurs constantly across the AI writing stack, that it has been hiding in plain sight because the field has no name for it, and that we can name it, taxonomize it, benchmark it, and detect it with the tools we already have. The remainder of Section 1 builds the context a reader needs to follow the technical sections.

2. How We Got Here: AI Mediation of Everyday Writing

2.1 The mechanics of phone keyboard autocorrect

Phone keyboards have intercepted what people type since the launch of the iPhone in 2007. Each keystroke is fed into a predictive system that decides three things: which key the user most likely intended (the touch model), which word they are most likely trying to spell (the language model), and whether to replace the typed word with its most probable correction (the correction policy). The decision happens in under fifty milliseconds, before the user sees the screen update.

Modern keyboards (Apple's UIKit autocorrect since iOS 8, Google's Gboard since 2016, SwiftKey since the Microsoft acquisition) use neural language models trained on billions of tokens of user text. These models are good at handling typos like *teh* to *the*, but they are statistical predictors. When the model assigns high probability to a correction that the user did not intend, the keyboard inserts it anyway. The user sees the wrong word for a fraction of a second, sometimes longer, and is left to either notice and undo it or press send.

Two properties of this system matter for our argument. First, the correction is *silent*: there is no warning that a substitution has happened, only the substituted word itself. Second, the correction *operates on the user's text* rather than generating new content. The keyboard rewrites what the user typed, and the resulting message is attributed to the user.

When the autocorrect system mishandles a non-Anglo name (say, replacing *Ayaan* with *Ian*, or *DaShawn* with *dash away*), it is doing the same thing it does on a typo, but the consequences are different. The "I Am Not a Typo" campaign in 2024 documented that Microsoft's English (UK) dictionary flagged 5,492 of the 13,532 unique names given to UK babies in 2021 as typos, with the rate disproportionately affecting names of Indian, African, Welsh, and Scottish origin [11]. Each flag triggers a potential silent correction. The cumulative effect, across billions of messages per day, is hard to quantify but easy to imagine.

2.2 From autocorrect to AI-mediated writing

Autocorrect is the simplest case of AI mediation in writing. There are now many richer cases. Gmail Smart Compose [9] suggests sentence completions as the user types; the user can accept the suggestion with a single keystroke. Google Smart Reply offers three pre-written responses to incoming emails. Apple Intelligence (introduced in iOS 18.1, October 2024) provides a Rewrite tool that takes selected text and produces formal, friendly, or concise variants. GitHub Copilot suggests code completions that include English-language comments and commit messages. Grammarly rewrites entire paragraphs at the user's request.

All of these systems share a structure. The user provides original text. The AI system produces a suggested rewrite. The user accepts or rejects the suggestion. When the user accepts (sometimes by reflex, sometimes without re-reading), the suggested rewrite becomes the user's message. The AI's output is now attributed to the human.

At the most ambitious end of this spectrum are *agent pipelines*: chains of language models that pass user text through several rewrite stages. A customer support agent might pass a user complaint through a classifier, a drafter, a tone-polisher, and a legal-review step before the final response goes back to the user. Frameworks like CrewAI, LangGraph, AutoGen, and ruflo make this kind of multi-hop pipeline easy to construct. The user-authored input has been edited by the time it reaches the recipient, sometimes substantially, with no clear record of what changed at each hop.

2.3 The shared failure mode

All of these systems can do exactly what the WhatsApp keyboard did to our opening message. They can change the polarity of a claim (turning a yes into a no, or vice versa). They can swap a content word for its opposite. They can strengthen a tentative commitment into a firm promise. They can change a quantifier from *some* to *all*. They can shift a register from casual to formal in ways that change the social meaning of a message. They can replace a name with one the dictionary recognizes. They can do all of this silently, with no warning to the user, and the resulting text is attributed to the human author.

Researchers have studied parts of this phenomenon under different names. Negation handling in natural language inference has decades of work behind it [1, 2]. Hallucination detection has become central to evaluating large language models [3, 4]. Sycophancy (the tendency of language models to agree with user-stated views) has been examined as a safety property [5]. The "I Am Not a Typo" campaign has drawn attention to name-mangling specifically. None of these threads, however, address the unified phenomenon: AI-mediated writing systems take user-authored text and produce variants whose pragmatic meaning differs from the user's intent. We call this *intent corruption*, and we argue it deserves its own taxonomy, benchmark, and detector.

2.4 Why this matters now

Three trends converge to make intent corruption a problem worth naming. First, AI-mediated writing is no longer a feature; it is the default. Gmail Smart Compose reportedly serves billions of suggestions per day. iOS autocorrect operates on every iPhone keyboard in the world. Smart Reply is now baked into LinkedIn, Slack, and every major messaging platform. Second, the systems have grown more aggressive: where 2015-era autocorrect would only fix obvious typos, 2024-era Apple Intelligence will rewrite an entire paragraph in a different tone with one click. Third, the rewrites are happening on text that increasingly carries weight: emails to customers, messages to family, code reviews, legal correspondence, medical advice.

When a Smart Compose suggestion subtly inverts a refusal into an agreement, or a Rewrite tool turns "I'll try to attend" into "I will attend," or an agent pipeline drafts an apology email that promises a refund the customer service team did not authorize, the consequences can be measurable. Yet there is no detection layer in any of these systems checking whether the rewrite preserved the user's intent. That is the gap we set out to close.

3. Contributions and Roadmap

This paper makes four contributions, each addressing one gap in the current literature.

A taxonomy. We propose twelve categories of intent corruption: polarity negation, polarity antonym, modal shift, commitment distortion, quantifier shift, temporal shift, subject-object swap, scope error, tone shift, hedging shift, identity corruption, and intent drift. Each category is defined in linguistic terms and grounded in pragmatic theory drawing on Horn [12], Searle and Vanderveken [13], Hyland [14], Halliday and Hasan [15], and Grosz and Sidner [16]. Section 4 lays out the taxonomy in detail.

A detector. We build SIL (Semantic Intent Layer), a detection system that operates across all twelve categories. SIL is a parallel-thinking ensemble that combines fast rule-based filters, a natural language inference cross-encoder (DeBERTa-v3-large MNLI [17]), and a large language model judge (Claude Haiku 4.5 [18]). The architecture is inspired by the parallel-thinking pattern used in the Databricks Genie system [19]. Section 5 describes SIL.

A benchmark. We construct ICOR-Bench, a 1,460-example labeled dataset drawn from four permissively licensed corpora: WANLI [6], PAWS [7], ParaDetox [8], and the Anthropic sycophancy evaluation set [26]. ICOR-Bench is the first benchmark designed specifically for intent-corruption detection. Section 6 describes its construction and SIL's performance on it.

An out-of-distribution evaluation. We validate SIL against the IteraTeR human-annotated revision corpus [20], using human revision-intent labels as ground truth. SIL's F1 drops from 93.9 percent on ICOR-Bench to 67.6 percent on IteraTeR, a 26-point gap. Section 7 reports this evaluation and Section 8 diagnoses the failure modes behind the gap. We argue this is the most important finding in the paper.

4. The ICOR Taxonomy

We define twelve categories of intent corruption. Each names a specific way an AI-mediated edit can change pragmatic meaning while leaving surface form roughly intact. Each is grounded in established theory of pragmatics, semantics, or discourse. We chose twelve because that is the number of categories at which the

boundaries between them stop overlapping for our annotators and the taxonomy still covers the in-the-wild cases we observed.

Polarity Negation. Insertion or removal of negation cues such that the polar valence of a proposition flips. Example: *I can take care of it* becomes *I can't take care of it*. This is the canonical autocorrect failure mode and the case in our opening anecdote. Grounded in classical logic and Horn's [12] treatment of negation as polarity reversal.

Polarity Antonym. Replacement of a content word with its lexical antonym. Example: *please approve the deployment* becomes *please reject the deployment*. Common when an LLM is given an instruction like "make this sound more decisive" and chooses a negative-valence verb. Related to Horn's lexical scales and to antonym pairs in WordNet [27].

Modal Shift. Change in modal strength along the certainty or obligation scale. Example: *this might be feasible* becomes *this must be feasible*. The two sentences make different commitments about the speaker's certainty. Grounded in Lassiter's treatment of graded modality [28] and Lyons's modal scales [29].

Commitment Distortion. Change in the illocutionary force of a commissive speech act. Example: *I'll try to send the report Friday* becomes *I will send the report Friday*. The rewrite has converted an attempt into a promise. Searle and Vanderveken's speech act theory [13] treats promising as a commitment to action; weakening or strengthening that commitment changes the act.

Quantifier Shift. Change along a quantifier scale. Example: *some users reported issues* becomes *all users reported issues*. The two sentences make logically different claims. Grounded in generalized quantifier theory and Horn's scalar implicature [12].

Temporal Shift. Change in temporal reference. Example: *let me know if you need it soon* becomes *let me know if you need it now*. The urgency conveyed differs. Grounded in tense and aspect semantics and Reichenbach's [30] event-time framework.

Subject-Object Swap. Reversal of argument roles such that the agent and patient exchange positions. Example: *I asked her to review the doc* becomes *she asked me to review the doc*. The sentence is still grammatical and on the same topic, but the responsibility has been transferred. Grounded in thematic role theory [31].

Scope Error. Misplacement of logical operators such as negation or quantifiers, producing non-equivalent sentences. Example: *not all reviewers approved* versus *all reviewers did not approve*. The two sentences have different truth conditions even though they share most of their words. Grounded in monotonicity reasoning [22] and De Morgan's laws.

Tone Shift. Change in register, politeness, or emotional valence. Example: *could you take another look at this when you have a moment* becomes *take another look at this now*. The literal request is the same, but the social meaning has changed from polite request to terse command. Grounded in Halliday's register theory [15] and the politeness corpus of Danescu-Niculescu-Mizil et al. [32].

Hedging Shift. Addition or removal of epistemic hedges (words like *I think*, *perhaps*, *somewhat*). Example: *the design needs revisions* becomes *I think the design needs revisions*, or vice versa. The level of speaker confidence expressed differs. Grounded in Hyland's hedging taxonomy [14] and Lakoff's hedges [33].

Identity Corruption. Mangling or replacement of personal names and identifiers, frequently affecting non-Anglo names. Example: *please loop in Ayaan* becomes *please loop in Ian*. This category captures the autocorrect-mangles-my-name phenomenon documented at scale by the "I Am Not a Typo" campaign [11].

Intent Drift. Cumulative pragmatic shift across multiple mediation hops, where each hop is small but the chain produces text that bears no resemblance to the user's original intent. This category is reserved for multi-step pipelines and for cases where the change is real but no single more-specific category fits. Grounded in Grosz and Sidner's [16] account of discourse intention structure.

We do not claim these twelve categories are exhaustive. We arrived at them through iteration: starting from observed autocorrect and Smart Compose failures, expanding through literature review of pragmatic phenomena, and refining through preliminary annotation rounds. A formal saturation study to test whether these twelve cover real-world corruption observations is future work.

5. SIL: The Detection System

SIL takes a pair of strings (the original user-authored text and the AI-suggested rewrite) and returns a per-category risk score together with one of three action labels: pass, warn, or block. The architecture is a parallel-thinking ensemble of three trajectories, each running a different subset of detectors. We describe how each trajectory works, then how their outputs are combined.

5.1 Trajectory 1: rule-based detectors

The fastest trajectory runs only deterministic rule-based detectors. For each category that has clean lexical signatures, we implement a regular-expression or lexicon-lookup check. The polarity-negation detector counts negation cues (*not*, *n't*, *never*, *no*) and flags pairs where the count differs by at least one. The polarity-antonym detector compares WordNet antonym sets between the two strings. The modal-shift detector ranks modals on a strength scale (*might* < *may* < *can* < *should* < *must*) and flags pairs that move across the scale. The hedging-shift detector counts hedge markers from Hyland's taxonomy. The full set of rules runs in under five milliseconds on commodity CPU.

Rules catch the obvious cases. They miss any corruption that is not expressible as a simple lexical pattern. They form the floor of SIL's capability, not the ceiling.

5.2 Trajectory 2: natural language inference

The second trajectory adds a natural language inference (NLI) cross-encoder. NLI is a well-studied task in NLP: given two sentences (a premise and a hypothesis), an NLI model classifies the relationship as entailment, neutral, or contradiction. We use MoritzLaurer's DeBERTa-v3-large fine-tuned on a combination of MNLI, FEVER, ANLI, LING, and WANLI [17], accessed through the HuggingFace Inference Providers router endpoint.

We run NLI in both directions. We feed the model (original, suggested) and (suggested, original) and combine the contradiction probabilities. Asymmetric entailment signal (where one direction entails but the other does not) contributes to scope-error and intent-drift detection. The full NLI trajectory completes in approximately 100 to 200 milliseconds per pair in our deployment.

5.3 Trajectory 3: LLM judge

The third trajectory adds a BGE-M3 embedding similarity check [34] and a large language model judge (Claude Haiku 4.5 [18] via the OpenRouter API). The LLM judge handles tone shifts, identity corruption, and subtle intent drift, which are the categories where lexical rules and NLI cross-encoders are weakest. The judge is constrained to produce strict JSON output, with a per-category risk score and a short rationale. The third trajectory completes in approximately one to three seconds per pair, dominated by the LLM API call.

5.4 Aggregation

The three trajectories run in parallel and produce per-category risk vectors. The aggregator combines them using soft-OR within each trajectory and a robust voting scheme across trajectories. When only one trajectory fires on a given category (for example, when only the LLM judge detects a tone shift that the rule-based detectors missed), that signal is trusted rather than diluted by the silence of the other trajectories.

The final risk score is computed as the maximum of weighted mean and weighted max across categories, with surface-specific weights. Edits on emails weight commitment shifts higher; edits on source code weight polarity and scope higher; edits on chat messages weight tone shifts higher. Action thresholds default to: pass if risk below 0.40, warn if between 0.40 and 0.70, block if above 0.70.

5.5 Deployment surfaces

SIL ships in three forms. A *ruflo plugin* registers a pre-edit hook that intercepts AI-suggested edits before they apply to user-authored text within the ruflo agent orchestrator. A *Model Context Protocol (MCP) server* exposes SIL as a set of tools callable by Claude Code, Cursor, and other MCP-aware hosts. A *standalone Fastify HTTP API* serves browser extensions, Slack bots, and other clients. All three share the same core detector pipeline.

6. ICOR-Bench: An In-Distribution Benchmark

ICOR-Bench is the first benchmark constructed specifically to evaluate intent-corruption detection across all twelve categories. This section describes how it was built, its known limitations, and SIL's performance on it.

6.1 Construction

We assemble 1,460 labeled tuples (original text, suggested rewrite, ICOR category, corruption-or-clean label) from four permissively licensed source corpora plus a small hand-written seed.

Source	License	Pairs	ICOR categories
WANLI [6]	CC-BY-4.0	405	polarity_negation, scope_error
PAWS-Wiki [7]	Free (Google)	204	subject_object_swap
ParaDetox [8]	CC-BY-4.0	204	tone_shift
Anthropic sycophancy [26]	CC-BY-4.0	406	commitment, intent_drift
Hand-written seed	—	31	all 12 categories (partial)

Clean negatives	mixed	210	true_paraphrase
Total		1,460	

Table 1. ICOR-Bench composition by source corpus. Seven categories have approximately 200 examples each. Five categories (polarity_antonym, modal_shift, quantifier_shift, temporal_shift, hedging_shift, identity_corruption) are under-sampled at four to five examples each, drawn from the hand-written seed only. Synthetic generation for the under-sampled categories is flagged as future work.

We use four source corpora because each one cleanly covers a different ICOR axis. WANLI is a 108,000-example natural-language-inference dataset created by Liu et al. [6] through worker-and-AI collaboration; its contradiction pairs map to polarity-negation and scope-error corruptions. PAWS-Wiki [7] consists of high-overlap non-paraphrases built by adversarial word scrambling on Wikipedia sentences; these map almost directly to subject-object swaps. ParaDetox [8] provides toxic-to-neutral paraphrase pairs which we invert to simulate tone-shift corruption. The Anthropic model-written-evals sycophancy subset [26] provides pairs where a language model yielded to a user-stated view; we use these as commitment-distortion and intent-drift examples.

6.2 SIL’s performance on ICOR-Bench

Metric	Value
Precision	90.2%
Recall	97.8%
F1	93.9%
Accuracy	89.0%
False-positive rate on clean class	63.3%
Latency, median	2.7 seconds
Latency, 99th percentile	6.6 seconds
Wall time for full run	76 minutes
API cost (OpenRouter)	\$0.88

Table 2. SIL performance on ICOR-Bench (N=1,460), with all three trajectories enabled and the default risk threshold of 0.40.

Recall is the headline number. SIL catches 97.8 percent of the corruption examples in ICOR-Bench. The cost is a high false-positive rate on the clean paraphrase class: SIL flags 133 of 210 clean paraphrase examples (63 percent) as corruption. Most of these false positives carry high confidence (51 of them have risk scores of exactly 1.0). The system is confidently wrong on natural paraphrases.

The mechanism, on inspection, is that the DeBERTa NLI cross-encoder treats substantial lexical change as evidence of contradiction even when meaning is preserved. WANLI entailment pairs and PAWS true paraphrases routinely involve sentence-level rewrites where the surface form changes but the underlying

meaning does not. NLI fires on these, the aggregator pushes risk past threshold, and SIL produces a false alarm.

7. IteraTeR Validation: Against Human Ground Truth

Performance on ICOR-Bench measures whether SIL agrees with the source-corpus labels we used to build the benchmark. It does not measure whether SIL agrees with humans on naturalistic AI-mediated edits. To measure that, we evaluate against IteraTeR [20], a corpus of Wikipedia revisions annotated by humans for revision intent.

7.1 What IteraTeR is

IteraTeR was developed by Du et al. [20] at Grammarly. It contains approximately 31,000 sentence-level revisions drawn from Wikipedia, ArXiv, and news sources. Each revision pair is annotated by linguistics-trained human raters with one of five intention labels: *fluency* (typo and grammar fixes), *coherence* (reorganization that preserves meaning), *clarity* (rewording for understandability), *style* (register changes), and *meaning-changed* (edits that change the literal meaning).

For our purposes, the value of IteraTeR is that the annotations are human-produced, third-party (not generated by us), and operate on a different distribution from ICOR-Bench. Wikipedia revisions are real edits made by real editors for real reasons. If SIL agrees with the human raters on these, that is meaningful evidence that the system catches what humans recognize as intent corruption.

7.2 Label mapping

We map IteraTeR's five-class labels to a binary task suitable for SIL evaluation. Meaning-changed maps to the corruption class. Fluency and coherence map to the clean class (the annotation guidelines describe both as intent-preserving). Clarity and style we report separately as ambiguous, because these are exactly the cases where pragmatic intent can shift without literal meaning changing, and where humans themselves disagree.

7.3 Results

Metric	Value	95% confidence interval
Precision	61.8%	[57.5, 66.4]
Recall	74.8%	[70.4, 78.6]
F1	67.6%	[64.0, 71.2]
Cohen's kappa	0.453	[0.403, 0.506]

Table 3. SIL performance on IteraTeR human-labeled revisions ($N=1,572$, of which 1,072 enter the binary task after excluding the ambiguous clarity and style categories). Bootstrap 95 percent confidence intervals from 1,000 resamples.

Cohen's kappa of 0.453 places SIL at *moderate* agreement with human annotators on Landis and Koch's standard scale [35]. F1 of 67.6 percent on the binary task is substantially below the 93.9 percent achieved on ICOR-Bench. Recall is better than precision: SIL catches roughly three of every four meaning-changes that humans flagged, but only 62 percent of its own flags are confirmed by human annotation.

Intent (human label)	N	SIL flag rate	What this means
fluency	400	8.0%	Correctly passes typo/grammar fixes
style	100	37.0%	Borderline; register shifts may matter
clarity	400	41.8%	Borderline; rewording for clarity
coherence	272	56.2%	Over-flags sentence reordering
meaning-changed	400	74.8%	Strong recall on real corruption

Table 4. SIL flag rate broken down by IteraTeR intent category.

The per-intent breakdown is more informative than the aggregate. SIL correctly passes 92 percent of fluency edits, the cleanest category. It correctly flags 75 percent of meaning-changed edits, the dirtiest. The middle three categories show the expected progression: clarity and style edits are genuinely ambiguous, and coherence edits are where the precision drain concentrates.

Of the 299 corruptions correctly identified by SIL, an important pattern emerges: 68.9 percent are classified by the catch-all *intent_drift* category, which is defined by low embedding-similarity rather than by specific linguistic phenomena. The remaining catches are distributed thinly across *polarity_negation* (8.4 percent), *scope_error* (7.4 percent), *identity_corruption* (6.7 percent), *modal_shift* (3.7 percent), *tone_shift* (2.0 percent), and others.

8. The Twenty-Six-Point Gap

SIL achieves F1 of 93.9 percent on ICOR-Bench and F1 of 67.6 percent on IteraTeR. The 26-point gap is the most important finding in this paper. We argue it is not a calibration issue, not a SIL-specific flaw, and not an artifact of dataset choice. It is a structural property of the synthetic-versus-naturalistic distinction in this domain. Any detector built on the techniques SIL uses (rules, NLI, embedding, LLM judge) is likely to exhibit a similar gap.

ICOR-Bench is built from corpora with deliberately constructed corruption pairs. WANLI contradictions, PAWS adversarial swaps, ParaDetox toxicity transfers, and Anthropic sycophancy probes all encode their target phenomena cleanly. A single linguistic axis changes per pair. The pairs are designed for diagnostic clarity, not for naturalness.

IteraTeR revisions look different. A typical meaning-changed Wikipedia edit might replace a date, swap a proper noun, refine a numerical claim, or rephrase a clause in a way that subtly alters scope. Multiple linguistic axes change together. Lexical overlap between original and revised versions is often high. The rule-based detectors look for cleanly isolated patterns and find nothing. The NLI cross-encoder sees substantial overlap and produces weak contradiction signal. Only the generic embedding-distance metric reliably triggers, which is why 69 percent of SIL's true positives on IteraTeR fall into the *intent_drift* catch-all.

Two specific failure modes compound the gap. **First, NLI over-fires on sentence reordering.** Coherence edits in IteraTeR frequently reorder clauses or sentences while preserving content. NLI cross-encoders are sensitive to word-order changes and produce contradiction signal even when meaning is identical. SIL flags 56 percent of IteraTeR coherence edits, and most of these are false positives.

Second, the category-specific detectors fail to fire on subtle naturalistic changes. The rule and lexicon detectors are calibrated on examples that contain clear polarity, modal, or quantifier markers. Real revisions often lack such markers; a meaning change might involve substituting one factual claim for another, with no linguistic cue visible to the detectors. The system catches that something has changed (via embedding distance) but cannot say what changed.

Both failure modes suggest concrete architectural improvements. The aggregator could penalize NLI's contradiction signal when the LLM judge agrees there is no semantic change. Category-specific detectors could be retrained on naturalistic revision corpora rather than synthetic perturbations. The `intent_drift` catch-all could be decomposed into finer subcategories. We leave these to follow-up work and discuss them in Section 10.

9. A Methodology Note: The WildChat Detour

We document a methodological dead end because it constitutes a usable artifact for subsequent researchers.

Our initial measurement plan was to apply SIL to WildChat-1M [36], a corpus of approximately 838,000 anonymized ChatGPT conversations released under a permissive license. The reasoning was that intent corruption should manifest in real AI-deployment traffic at measurable rates, and WildChat offered the largest available sample of such traffic.

We piloted SIL on $N=1,000$ (user-prompt, assistant-response) pairs from WildChat. SIL flagged 96.9 percent of pairs as corruption, with 53.9 percent of flags routed to `polarity_negation`. The result is methodologically broken, not a property of ChatGPT. SIL is built to evaluate paraphrase-class rewrites, where the suggested output preserves the meaning of the original with modified surface form. Prompt-response pairs have no such entailment relationship: the response answers or expands the prompt rather than paraphrasing it. The NLI cross-encoder, fed two lexically dissimilar strings with no meaning-preservation relationship to evaluate, fires its contradiction signal almost universally.

A second pilot, with a regex filter to extract rewrite-style turns ("polish this:", "rewrite the following:"), yielded approximately 33 percent extraction precision due to semantic ambiguity in user prompts. Phrases like *polish my idea: write me a paragraph about X* match the rewrite regex but are in fact content-generation requests. We abandoned the WildChat plan in favor of the IteraTeR evaluation described in Section 7.

We report this in detail because it constitutes a finding: intent-corruption detectors evaluated on prompt-response pairs from production chat corpora will produce systematically misleading flag rates. Future work in this area should use rewrite-class corpora (IteraTeR, arXivEdits, NewsEdits, GYAFC) or filtered subsets that isolate genuine rewrite turns. The design surface of any intent-corruption detector is paraphrase-class edits, and evaluation must respect that surface.

10. Limitations and Ethical Considerations

We list the limitations of this work explicitly so they can be addressed in future work and so reviewers have a clean account of what we did and did not establish.

Five categories are under-sampled. In ICOR-Bench, `polarity_antonym`, `modal_shift`, `quantifier_shift`, `temporal_shift`, `hedging_shift`, and `identity_corruption` each have only four or five examples, drawn from the

hand-written seed. The reported 100 percent accuracy on these categories is not statistically meaningful. We have a synthetic generation harness that can address this gap using WordNet antonyms, the Lassiter modal-strength scale [28], Horn quantifier scales [12], TIMEX3 expressions, Hyland hedge lists [14], and Census and WikiData name lists. Running these generators and validating the output is the most immediate piece of follow-up work.

IteraTeR annotations are themselves human-produced labels. We did not re-annotate IteraTeR examples to confirm the original labels. Some apparent SIL errors may be IteraTeR annotation noise rather than model error. A planned manual audit of 30 false negatives and 30 false positives will bracket the share of disagreement attributable to annotation noise versus model error.

All experiments are English-only. PAWS-X and multilingual NLI corpora exist and would enable a multilingual extension. The identity-corruption category is particularly relevant in non-English contexts (the "I Am Not a Typo" findings are explicitly about names of South Asian, African, Welsh, and Scottish origin), and we expect cross-lingual evaluation to be the most informative next step in this category.

No user study has been run. The claim that SIL's interventions help users recover from intent corruption is currently theoretical. We have not measured whether users prefer SIL-corrected outputs to uncorrected AI rewrites, nor whether exposure to SIL warnings reduces user-encountered corruption in real writing tasks. A within-subjects user study is planned for the follow-up venue submission.

Ethical considerations. The screenshot in Figure 1 is reproduced with the explicit written consent of the friend named in it. No personally identifiable information beyond a first name (used with consent) is reproduced. All source corpora are released under CC-BY-4.0, CC-BY-SA-3.0, Open Data Commons, or equivalent permissive licenses, and we redistribute only the derivative ICOR-Bench under CC-BY-SA-4.0 in compliance with the share-alike chain. In deployment, intent-corruption detectors could introduce friction for users whose writing involves heavy editing (translators, language learners, writers in second languages); we recommend deploying in warn mode rather than block mode in such contexts, with per-population validation before shipping. The system's bias profile on non-Anglo names has been designed for but not empirically measured; this is a priority for the follow-up.

11. Conclusion and Future Work

Intent corruption is the silent change in pragmatic meaning that AI writing systems introduce when they edit user-authored text. The phenomenon has been operating in production at scale for over a decade and has been hiding in plain sight because the field has not given it a name. This paper provides the first taxonomy, benchmark, and detector for it. SIL achieves F1 of 93.9 percent on the synthetic ICOR-Bench and 67.6 percent on naturalistic IteraTeR revisions, with Cohen's kappa of 0.453 against human annotators. The 26-point gap between synthetic and naturalistic evaluation reveals that benchmark performance in this domain systematically overstates real-world detection capability.

Three follow-up papers extend this work naturally. The first is the architectural improvement paper: paraphrase-aware NLI reweighting, naturalistic-revision-trained category detectors, finer-grained intent_drift decomposition, and the threshold and trajectory ablations whose infrastructure we have shipped but not yet run. The second is the cascade dynamics paper: how does intent corruption compound across multiple AI mediation hops, and what growth curves describe drift in homogeneous versus heterogeneous pipelines? The

third is the deployment study: a controlled user evaluation of whether SIL's interventions actually help users recover from AI-induced intent corruption in real writing tasks.

We close where we started, with the WhatsApp screenshot from May 17, 2026. The phone keyboard inserted an apostrophe and a letter t. The friend believed the opposite of what was meant for forty-seven minutes. There was no detection layer in the loop, and there should have been. This paper is one step toward building that loop, in software and as a research discipline. The work begins, as it often does, with someone noticing.

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Appendix A: Reproducibility

All code, the ICOR-Bench JSONL file, and the IteraTeR evaluation scripts are released at github.com/plutaslab/ruflo-sil under MIT license for code and CC-BY-SA-4.0 for data. The repository contains the complete TypeScript implementation of SIL with three constructor paths (SIL.local for offline mock-based operation, SIL.hosted for the HuggingFace plus OpenRouter stack used throughout this paper, and SIL.modal for self-hosted DeBERTa NLI), the Pillar A extraction recipes for WANLI, PAWS, ParaDetox, and the Anthropic sycophancy subset, the IteraTeR evaluation script with stratified sampling and bootstrap confidence interval computation, and a Fastify HTTP API server exposing the scoring endpoints.

Software versions used in the reported experiments: Node 20.19.5, TypeScript via tsx, Python 3.13.5, HuggingFace datasets 4.8.4, DeBERTa-v3-large-MNLI-FEVER-ANLI-LING-WANLI accessed via the HuggingFace Inference Providers router, BAAI/bge-m3 accessed via the same router, and

anthropic/claude-haiku-4.5 accessed via OpenRouter. Full reproduction of the ICOR-Bench evaluation takes approximately 76 minutes of wall time and costs approximately \$0.88 in OpenRouter credit. Reproduction of the IteraTeR evaluation takes approximately 27 minutes and costs approximately \$0.94. Bootstrap confidence intervals use 1,000 resamples per estimate.